

COMP1531

Software Engineering Fundamentals

Week 9

Requirements - Validation

In This Lecture

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Why?

- It's important to understand the difference between verification and validation

What?

- Validation
- User Acceptance Testing

Let's Compare

*ISO/IEC/IEEE 29148:2018

Verification

- Verification in a system life cycle context is a set of activities that compares a product of the system life cycle **against the required characteristics for that product**. This may include, but is not limited to, specified requirements, design description and the system itself.

Validation

- Validation in a system life cycle context is a set of activities ensuring and gaining confidence that a system is **able to accomplish its intended use, goals and objectives**.

Let's Compare

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Verification

- System will be built right

Validation

- Right system will be built

Why Do We Validate?

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- The aim is to pick up any problems before resources are committed to addressing the requirements.
- Requirements validation is concerned with the process of examining the requirements document to ensure that it defines the right software (that is, the software that the users expect).

How Do We Validate?

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- Requirements Reviews
- Prototyping
- Acceptance Tests

User Acceptance Testing

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- [ISTQB](#) defines acceptance testing as **formal testing with respect to user needs, requirements, and business processes** conducted to determine whether or not a system satisfies the acceptance criteria and to enable the user, customers or other authorized entity to determine whether or not to accept the system.
- This testing is always black-box, and is typically tested on the customer or user themselves.
- Can either be performance based (trying to reach specific outcome) or stress testing. Sometimes you'll hear this testing environment be referred to as UAT.

Feedback

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